A

Threat Model: N-Agent Mesh

Adversary
no model or key access

Write channels
User query · Tool / API
Peer agent · Self-written

poisoned entry

Agent 1 — false fact — Memory store
valid signature, but the agent was misled

forward, hop 1

Agent 2 — copy — Memory store
copies reach neighbors

summarize, hop 2

Agent 3 — summary copy — Memory store
summaries inherit poison

via 4…N−1

Agent N — reworded copy — Memory store
reworded copy misleads later

Agentic IoT applications
IoT Agents with Shared Memory

- **Connected vehicles** — route and traffic memory
- **UAV fleets** — mission and map memory
- **Industrial IoT** — maintenance logs
- **Smart city** — gateway service memory

B

Proposed *EngraMesh* Framework

New entry from any channel

Runs locally on the edge devices (e.g., Raspberry Pi and Jetson Orin Nano)

Layer 1
Signed Envelope

Memory entry
- Writer ID
- Write channel
- Parent IDs
- Trust label, expiry
- Content hash

Ed25519 signed

Merkle log · **Verify**

Layer 2
Lightweight Detector

Stage 1: Text classifier

Entry text → MiniLM embedding → MLP classifier → Content score

Stage 2: Fusion

Provenance — channel, reputation, lineage

Network — hops, holders, spread

Fusion (LR)

Layer 3
Accept, Quarantine, or Recall

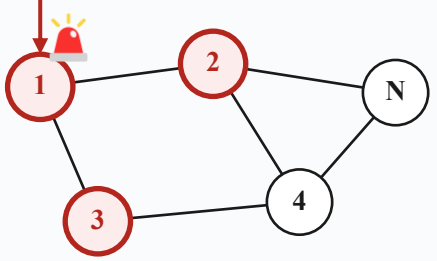
Risk score *s*

- $s < 0.3$ — **Accept** — Kept in memory
- $0.3 \le s < 0.7$ — **Quarantine** — Held for review
- $s \ge 0.7$ — **Recall** — Removed and sent to agents

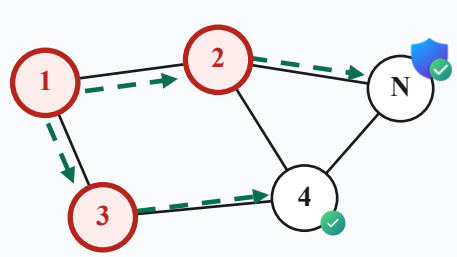
RECALL(entry, lineage)
signed, also removes near-copies

C

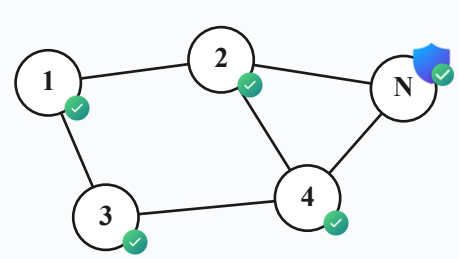
Network-Wide Recall
Removing the poisoned entry across the mesh



Agent 1 detects the poisoned entry when it is used



Signed RECALL hops along other agents and removes near-copies



Mesh cleared and late copies blocked

Legend
- holds the poison
- clean agent
- RECALL hop
- flags the entry
- blocks late copies
- inside a device
- between agents